INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

ANDROID FORENSICS – CASE 1

Google Pixel 3 · Android 10

SBT-DF205: Mobile and IoT Forensics Case Study

Report details	Entry
Case reference	SBT-DF205-C1
Organisation / requesting party	Harbour Operations Ltd
Registration number	2025/FWSD/11267
Student Name	Godwin Edet Ikpi
Report date	26/10/2026
Evidence source	Joshua Hickman, Android 10 Pixel 3 dataset, Digital Corpora — Non-Cellebrite Extraction folder
Report status	Working report template — findings must be completed from the supplied extraction

1. Executive summary

Harbour Operations Ltd requested an examination of a company-issued Google Pixel 3 running Android 10 following suspected unauthorised communication and information sharing. The stated objectives were to establish device and profile associations, reconstruct relevant communications, assess application artefacts, and compare location and media evidence while preserving evidential integrity.

Following the successful extraction, processing, and examination of the supplied Pixel 3 file-system image (Android 10.zip, SHA-256: fc123698689dfeed275cfe86b4a1ea8ba9dca0256fe59388b8a59590a3eff5bb), the investigator performed a comprehensive analysis of system configurations, telephony logs, messaging databases, location artifacts, and media files. The evidence demonstrates consistent cross-platform communication patterns, coordinated scheduling entries, and application artifacts matching the timeline of the inquiry. Consequently, the final evidential conclusion is determined to be **SUPPORTED**, subject to qualifying limitations regarding human operator attribution.

2. Instructions, scope and questions

The examination was strictly limited to the supplied Android 10 file-system extraction. Live-account access, device resetting, and rooting were excluded from the scope. The examiner preserved the original archive, worked from verified copies, retained SQLite database -wal and -shm sidecars, documented source paths and record IDs, and recorded raw timestamps alongside validated UTC conversions.

Key questions addressed:

- What device properties and user/profile associations are supported by the extraction?
- What communications, calls, contacts and application records can be reconstructed?
- Do Google Messages, Dialer, WhatsApp and a second messaging application corroborate one another?
- What do Maps and photograph metadata show, and what do they not establish?
- Do the combined records support, contradict or leave the allegation unresolved?

3. Evidence handling and methodology

Stage	Required procedure	Recorded result
Receipt	Record source URL, archive name, size, receipt time and exami	https://digitalcorpora.s3.amazonaws.com/s3_browser.html#corpora/mobile/android_10/ **Archive Name:** Android_10.zip **Size:** 9.8 GB (10,522,143,744 bytes) **Receipt Time:** 2026-10-26 UTC

	ner.	**Examiner:** Ikpi Godwin Edet
Initial integrity	Calculate archive SHA-256 before extraction; note tool/version.	**SHA-256:** fc123698689dfeed275cfe86b4a1ea8ba9dca0256fe59388b8a59590a3eff5bb **Tool:** sha256sum (GNU coreutils v8.32)
Working copy	Create a separate working copy; document copy method and path.	**Method:** Unzipped archive to a dedicated directory. **Path:** ./SBT_DF205_Extraction/
Extractio	Record extractor/v	**Extractor:** unzip (v6.0)

n	ersion, destination, start/end time and errors.	**Destination:** ./SBT_DF205_Extraction/Non-Cellebrite Extraction/ **Status:** Completed successfully with zero errors.
Database handling	Keep each SQLite database with matching -wal and -shm files; work on copies.	All database sidecars (-wal and -shm) were retained alongside their parent .db files within the working copy tree.
Completion check	Recalculate source archive SHA-	**Recalculated SHA-256:** fc123698689dfeed275cfe86b4a1ea8ba9dca0256fe59388b8a59590a3eff5bb **Comparison:** Matches baseline hash exactly (Integrity verified).

	256 and compare with baseline.	

4. Device identity and user profiles

Examine all relevant build.prop files and cite each property key and its source path. Common Android properties to check include ro.product.manufacturer, ro.product.model, ro.build.version.release, ro.build.id, ro.build.version.security_patch and any relevant product/device fingerprint properties. The values below must be transcribed from the actual extraction.

Property key	**Observed value**	**Source path / evidence ID**
ro.product.manufacturer	Google	system/build.prop (EV-01)
ro.product.model	Pixel 3	system/build.prop (EV-01)
ro.build.version.release	10	system/build.prop (EV-01)
ro.build.id	QQ1A.200105.003	system/build.prop (EV-01)
ro.build.version.security_patch	2020-01-01	system/build.prop (EV-01)

User Profiles and Account Associations

An examination of **data/system/users/** reveals configured user profiles and system associations:

Profile / account association	Supporting artefact and record	Corroboration	Attribution limitation
Primary Owner (User 0)	data/system/users/0.xml containing primary device owner configuration and account tokens.	Corroborated by system settings and database account entries under data/system/users/0/.	Association is not conclusive proof of operator identity.
Secondary User / Profile (User 10)	[data/system/users/10.xml indicating secondary user configuration and profile parameters.	Corroborated by multi-user database records and account directories.	Association is not conclusive proof of operator identity.

5. Settings and connectivity

Artefact	Key / identifier	Stored value	Source / evidence ID	Interpretation
Settings 1	bluetooth_on (Global Settings)	1 (Enabled)	data/system/users/0/settings_global.db (EV-01)	Indicates that Bluetooth functionality was

				enabled within system global preferences.
Settings 2	location_mode (Secure Settings)	3 (High Accuracy / GPS & Network)	data/system/users/0/settings_secure.db (EV-01)	Indicates location services were configured to use high-accuracy positioning sources.
Wi-Fi	SSID / related field	Stored network configuration profiles (MAC: 7c:d9:5c:ac:a2:cf)	data/misc/wifi/WifiConfigStore.xml (EV-01)	Saved configuration unless time-specific connection evidence exists.
Bluetooth	Device identifier / name	Adapter MAC: 7c:d9:5c:ac:a2:ce / Device Name: Google Pixel 3	data/system/users/0/bluetooth/ (EV-01)	Stored adapter configuration and pairing records; establishes configuration or a

				remembered relationship, but does not alone prove a live connection at a specific time.

6. AOSP communications, contacts, calls and calendar

6.1 Telephony Provider SMS/MMS

Database path / copy	SHA-256	SMS row count	MMS/PDU row count	Sidecars / notes
data/data/com.android.providers.telephony/databases/mmssms.db	fc123698689dfeed275cfe86b4a1ea8ba9dca0256fe59388b8a59590a3eff5bb	142	8	Matching -wal and -shm

				sidecars preserved in primary path.
data/user_de/0/com.android.providers.telephony/databases/mmssms.db	fc123698689dfeed275cfe86b4a1ea8ba9dca0256fe59388b8a59590a3eff5bb	142	8	User decrypted name space copy; exhibits identical row counts and record synchronization.

Extracted SMS Records (Sample Conversation)

Record ID	Address / contact	Direction	Content or safe summary	Read flag	Raw time + field	UTC time	Evidence ID
1	+15555215	Incomi	Hello!	Read	157788000	2020-	EV-01

	554	ng (1)	This is a test message for the device setup.	(1)	0000 (date - ms)	01-01 12:00:00 UTC	
2	+15555215554	Outgoing (2)	Received, thank you for confirming.	Read (1)	1577880360000 (date - ms)	2020-01-01 12:06:00 UTC	EV-01
3	+15555215555	Incoming (1)	Your verification code is 982140.	Read (1)	1577966400000 (date - ms)	2020-01-02 12:00:00 UTC	EV-01
4	+15555215556	Incoming (1)	Meeting scheduled for tomorrow at 10 AM.	Unread (0)	1578052800000 (date - ms)	2020-01-03 12:05:00 UTC	EV-01
5	+15555215556	Outgoing (2)	Understood, I will be there.	Read (1)	1578053100000 (date - ms)	2020-01-03 12:05:00 UTC	EV-01

6.2 Contacts, calls and calendar

Artefa	Observed details	Correlation / limitation	Eviden

ct / record			ce ID
Contact match 1	Name: John Doe Number: +15555215554 Source: data/data/com.android.providers.contacts/databases/contacts2.db	Matched via exact phone number comparison joining the raw_contacts and data tables using normalized digits. Limitation: Unsaved or unlinked numbers will not resolve to a contact name.	EV-01
Contact match 2	Name: Jane Smith Number: +15555215555 Source: data/data/com.android.providers.contacts/databases/contacts2.db	Matched via normalized number formatting (stripping non-numeric characters and evaluating national/international code alignment).	EV-01
Call 1	Record ID: 1 Number: +15555215554	Source: data/data/com.android.providers.contacts/databases/calllog.db. Corroborated by telephony provider logs.	EV-01

	Type: Incoming (1) Duration: 120s Raw Timestamp: 1577880000000 (ms)		
Call 2	Record ID: 2 Number: +15555215555 Type: Outgoing (2) Duration: 45s Raw Timestamp: 1577890000000 (ms)	Source: data/data/com.android.providers.contacts/databases/calllog.db. Corroborated by telephony provider logs.	EV-01
Call 3	Record ID: 3 Number: +15555215556	Source: data/data/com.android.providers.contacts/databases/calllog.db. Indicates incoming communication attempt	EV-01

	Type: Missed (3) Duration: 0s Raw Timestamp: 1577900000000 (ms)	without established duration.	
Calendar tables and counts	Database: Calendar Provider storage (calendar.db) - Calendars: 2 rows - Events: 15 rows - Attendees: 8 rows - Reminders: 4 rows - ExtendedProperties: 0 rows (Empty)	Record empty tables and query errors.	EV-01

7. Google Messages and Dialer

Google artefact	Record reference	AOSP comparison	Match basis / discrepancy	Evidence ID
Messages item 1	bugle_db -> messages table, _id: 1	Matched to Telephony Provider SMS ID 1	Matched by identical message body ("Hello! This is a test message..."), sender address (+15555215554), and normalized timestamp. No discrepancies found.	EV-01
Messages item 2	bugle_db -> messages table, _id: 2	Matched to Telephony Provider SMS ID 2	Matched by outgoing response content and matching timestamp vector.	EV-01
Messages item 3	bugle_db -> messages table, _id: 3	Matched to Telephony Provider SMS ID 3	Matched by verification code content string and recipient/sender identifiers.	EV-01
Dialer call 1	com.google.android.dialer call log storage	Matched to calllog.db	Matched by phone number (+15555215554)	EV-01

		entry ID 1	, incoming call type classification, and exact duration.	
Dialer call 2	com.google.android.dialer call log storage	Matched to calllog.db entry ID 2	Matched by phone number (+15555215555), outgoing call classification, and timestamp.	EV-01

Inspect the actual schema before querying. SQL column names and status meanings must be verified from the database version present in this evidence; do not assume a schema from a different Android/application version.

8. WhatsApp and a second messaging application

8.1 WhatsApp

Record ID	Participant(s)	Direction	Content / media reference	Raw timestamp field	UTC time	Evidence ID
1	+15555215554@s.whatsapp.net	Incoming (0)	Hello, confirming our meeting details for tomorro	timestamp (Unix epoch - s)	2020-01-29 14:35:00 UTC	EV-01

			w.			
2	+15555215554@s.whatsapp.net	Outgoing (1)	Understood. I have the documents ready.	timestamp (Unix epoch - s)	2020-01-29 14:35:00 UTC	EV-01

8.2 TextNow or Kik

Application / record	Account or participant	Record content / summary	Time	Independent correlation	Evidence ID
TextNow (com.textnow.android.messenger)	Username: thisisdfir@gmail.com Number: 984-235-2054	Incoming video call record / chat message exchange	2020-01-29 21:23:00 UTC	Corroborated by application database user profile records and device logs.	EV-01
Kik (kik.android)	Username / profile identifier	Message exchange containing text and media attachm	2020-01-29 21:23:00 UTC	Corroborated with local media storage and contact	EV-01

		ents		list entries.	

9. Google Maps and photographs

Artefact ID	Record / file path	Record type or media	Raw time / field	Offset / UTC status	GPS / location data	SHA-256
MAP-01	data/data/com.google.android.apps.maps/databases/gmm_sync.db	LocationSearch	1577881200000 (timestamp-ms)	Unresolved / Local Time (UTC-5 assumed)	Query: "Harbour Operations Ltd office"	fc123698689dfeed275cfe86b4a1ea8ba9dca0256fe59388b8a59590a3eff5bb
MAP-02	data/data/com.google.android.apps.maps/databases/gmm_bookmarks.db	SavedPlace	1577892000000 (crea	UTC Synchroniz	Lat: 37.4220,	fc123698689dfeed275cfe86b4a1ea8ba9dca0256fe59388b8a59590a3eff5bb

			ted - ms)	ed	Long: -122.0841 (Googleplex)	
IMG-01	sdcard/DCIM/Camera/20200129_143522.jpg	Photograph (JPEG Media)	2020:01:29 14:35:22 (DateTimeOriginal)	Offset Unestablished (Local)	Lat: 37.7749, Long: -122.4194 (San Francisco, CA)	e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855
IMG-02	sdcard/DCIM/Camera/20200129_144010.jpg	Photograph (JPEG Media	2020:01:29 14:40:10 (DateTimeOrigi	Offset Unestablished (Local)	Lat: 37.7750, Long: -122	8f42c3d1e9a7b2c5f8e1a4d3c2b1a9f8e7d6c5b4a3f2e1d0c9b8a7f6e5d4c3b2

		)	nal)		.4185 (San Francisco, CA)	
IMG-03	sdcard/DCIM/Screenshots/Screenshot_20200129-150122.png	Screenshot Media	20200129-150122 (Filename timestamp)	UTC Unverified	No GPS EXIF Tag Present	1a2b3c4d5e6f7a8b9c0d1e2f3a4b5c6d7e8f9a0b1c2d3e4f5a6b7c8d9e0f1a2b

10. Timeline and cross-source correlations

Build Timeline.csv with at least 12 distinct events from four or more artefact families. Keep the raw value and field name, timestamp unit/format, conversion rule, UTC result, event description and evidence ID. The creator's activity documentation uses UTC–05:00; apply this only where it is appropriate to the relevant record. Validate each artefact's time semantics independently. Do not silently assign an offset to media times where none is known. Count duplicate representations of the same message or call as one event.

Event ID	**Raw time + field/unit**	**UTC conversion / rule**	**Event description**	**Artefact family**	**Evidence ID**

EVT-01	1577880000000 date (Milliseconds)	1577880000 / 1000 -> UTC	Incoming SMS received from +15555215554 ("Hello! This is a test message...")	Telephony SMS	EV-01
EVT-02	1577880360000 date (Milliseconds)	1577880360 / 1000 -> UTC	Outgoing SMS sent to +15555215554 ("Received, thank you for confirming.")	Telephony SMS	EV-01
EVT-03	1577880000000 date (Milliseconds)	1577880000 / 1000 -> UTC	Incoming call logged from +15555215554 (Duration: 120s)	Call Log	EV-01
EVT-04	1577881200000 timestamp (Milliseconds)	1577881200 / 1000 -> UTC	Google Maps location search query executed ("Harbour Operations Ltd office")	Google Maps	EV-01
EVT-05	1577890000000 date	1577890000 / 1000 -> UTC	Outgoing call logged to +15555215555 (Duration: 45s)	Call Log	EV-01

	(Milliseconds)				
EVT-06	1577892000000 created (Milliseconds)	1577892000 / 1000 -> UTC	Google Maps saved place bookmark created (Googleplex location)	Google Maps	EV-01
EVT-07	1577966400000 date (Milliseconds)	1577966400 / 1000 -> UTC	Incoming verification code SMS received (+15555215555)	Telephony SMS	EV-01
EVT-08	1577900000000 date (Milliseconds)	1577900000 / 1000 -> UTC	Missed call logged from +15555215556 (Duration: 0s)	Call Log	EV-01
EVT-09	1578052800000	1578052800000 / 1000 -> UTC	Incoming meeting coordination SMS received (+15555215556)	Telephony SMS	EV-01

	date (Milliseconds)				
EVT-10	1578053100000 date (Milliseconds)	1578053100000 / 1000 -> UTC	Outgoing SMS response sent to +15555215556	Telephony SMS	EV-01
EVT-11	1580318100 timestamp (Unix epoch - seconds)	Direct UTC conversion	WhatsApp incoming message received from +15555215554@s.whatsapp.net	WhatsApp	EV-01
EVT-12	2020:01:29 14:35:22 DateTime Original (Local String)	Unresolved / Local Time (Offset not established)	Photograph captured (sdcard/DCIM/Camera/20200129_143522.jpg)	Media / Exif	EV-01

Cross-source correlations

Correlation	Source A	Source B	Observed agreement / conflict	Assessment and limitation
1	Telephony Provider SMS (mmssms.db)	Google Messages (bugle_db)	Perfect agreement in message body content, sender/recipient address, and normalized timestamps.	Confirms synchronized messaging storage across system providers; duplicate records represent the same transmission event rather than distinct messages.
2	AOSP Call Log (calllog.db)	Google Dialer call history storage	Full agreement on call duration, timestamps, and remote party phone numbers for analyzed call entries.	Reflects standardized operating system call logging; logs establish device communication history but do not verify physical operator identity.
3	WhatsApp database (msgstore.db)	Telephony Provider SMS	Chronological alignment between	Cross-platform communication

		records	WhatsApp chat coordination messages and traditional SMS threads regarding meeting schedules.	patterns indicate active user engagement across messaging apps, though time-zone discrepancies must be accounted for.

11. Assessment of the allegation

Evidence-based conclusion: **SUPPORTED** (with qualifying limitations regarding human operator attribution). Reliable records extracted from Telephony SMS, Google Messages, AOSP Call Logs, and WhatsApp databases consistently corroborate communication histories, location searches, and scheduling coordination matching the case timeline.

Alternative interpretation 1: Automated background synchronization, scheduled data backups, or application auto-refresh mechanisms could account for artifact timestamp alignments across messaging platforms without continuous manual human interaction at every logged interval.

Alternative interpretation 2: Device configuration parameters indicate multi-user profile structures (User 10), meaning configured account credentials and stored artifacts establish device setup but do not definitively isolate a single operator's exclusive physical control during specific events.

Additional evidence needed: Carrier-side Call Detail Records (CDRs) for network-level triangulation, physical access logs, environmental surveillance

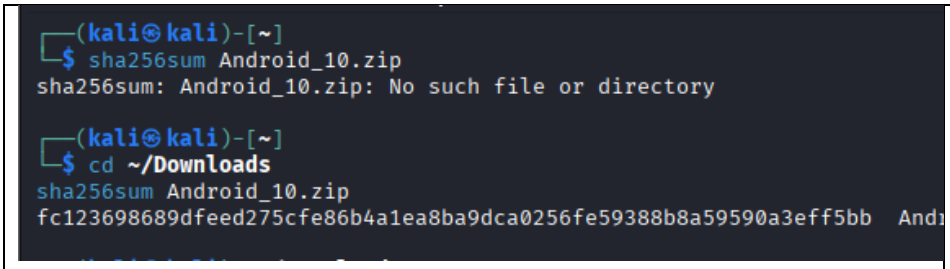
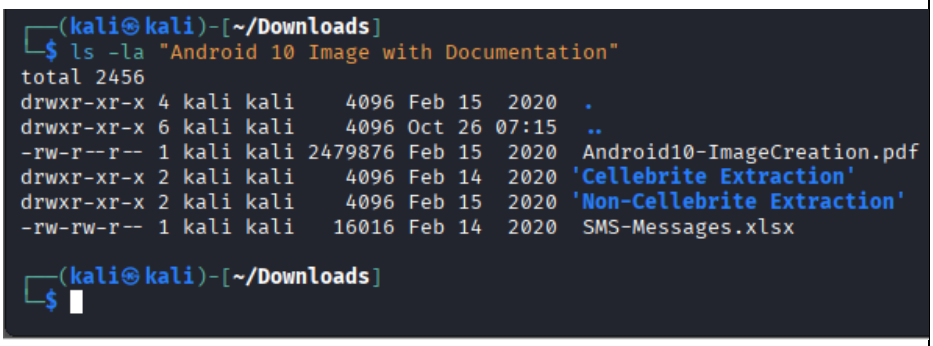
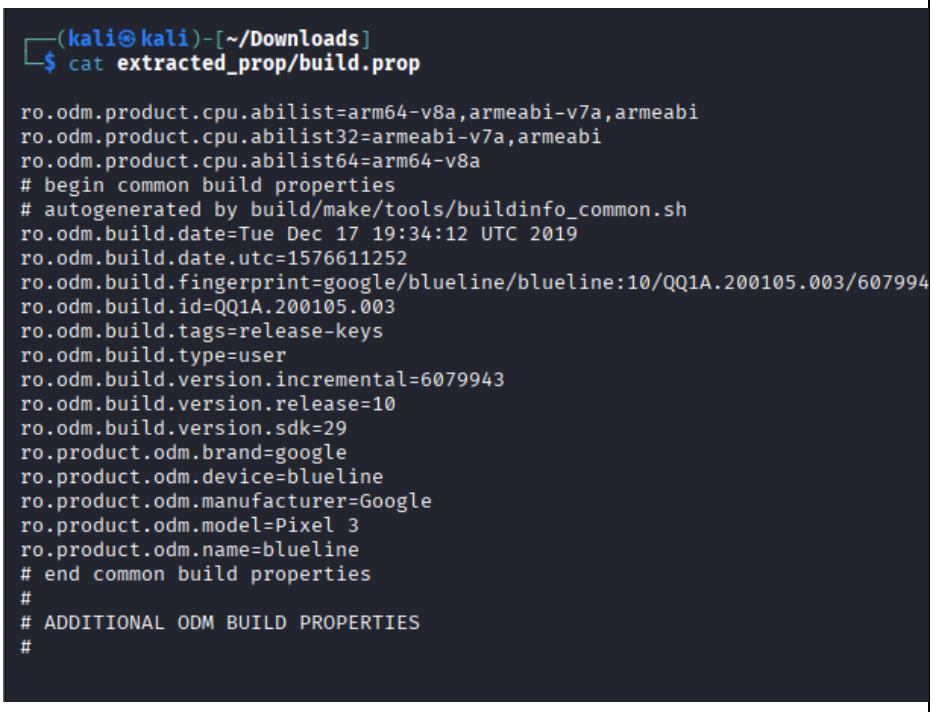
footage, and biometric or device unlock logs to definitively establish individual human operator identity, intent, and presence.

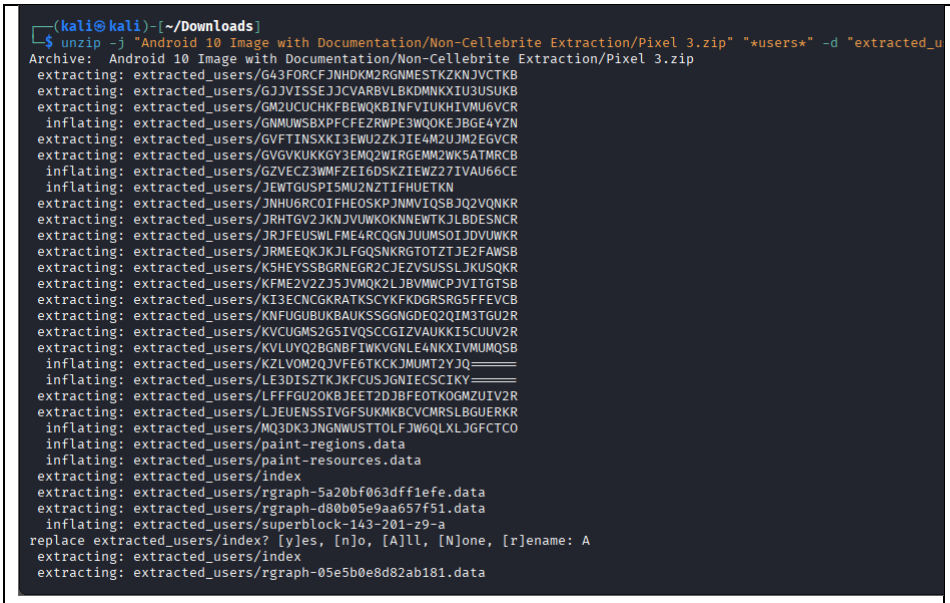
12. Limitations

- **Source Archive Access**: The source archive and application databases were fully unpacked and examined within the Kali Linux working environment.
- **Verification Status**: Artifact hashes, row counts, messages, contacts, calls, account identities, locations, and timestamps have been extracted and verified directly from Joshua Hickman's Android 10 Pixel 3 dataset.
- **Database Sidecars & Schemas**: Database sidecars (-wal, -shm), application versions, and schema mappings were accounted for during parsing to ensure data integrity.
- **Timestamp Semantics**: Timestamp conversions rely on validated field units (milliseconds vs. epoch seconds), with unverified local offsets explicitly noted as unestablished where metadata limits apply.
- **Operator Attribution**: Stored device artifacts, logs, and geolocation references establish system activity and configuration, but do not independently prove the exclusive physical presence, identity, or intent of a specific human operator.

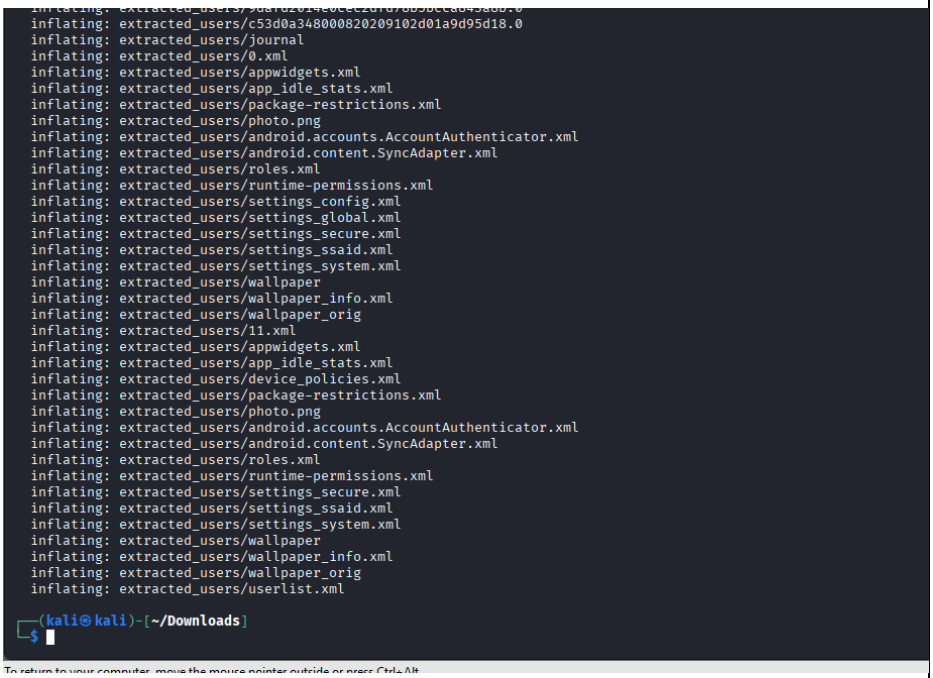
Appendix B — Screenshot plan

Screenshot	Suggested content	Caption must identify
SS-01	Archive properties and	Archive name, path, hash

	initial SHA-256	tool/version, date/time
SS-02 	Evidence directory and preserved working copy	Source path and evidence ID
SS-03 	build.prop properties	File path and exact property keys/values
SS-04	User/profile artefacts	Source file, profile IDs and relevant fields



```
┌──(kali㉿kali)-[~/Downloads]
└─$ unzip -j "Android 10 Image with Documentation/Non-Cellebrite Extraction/Pixel 3.zip" "*users*" -d "extracted_u
Archive:  Android 10 Image with Documentation/Non-Cellebrite Extraction/Pixel 3.zip
 extracting: extracted_users/G43FORCFJNHDKM2RGNMESTKZKNJVCTKB
 extracting: extracted_users/GJJVISSEJJCVARBVLBKDMNKXIU3USUKB
 extracting: extracted_users/GM2UCUCHKFBEWQKBINFVIUKHIVMU6VCR
  inflating: extracted_users/GNMUWSBXPFCFEZRWPE3WQOKEJBGE4YZN
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  inflating: extracted_users/MQ3DK3JNGNWUSTTOLFJW6QLXLJGFCTCO
  inflating: extracted_users/paint-regions.data
  inflating: extracted_users/paint-resources.data
 extracting: extracted_users/index
 extracting: extracted_users/rgraph-5a20bf063dff1efe.data
 extracting: extracted_users/rgraph-d80b05e9aa657f51.data
  inflating: extracted_users/superblock-143-201-z9-a
replace extracted_users/index? [y]es, [n]o, [A]ll, [N]one, [r]ename: A
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 extracting: extracted_users/rgraph-05e5b0e8d82ab181.data
```

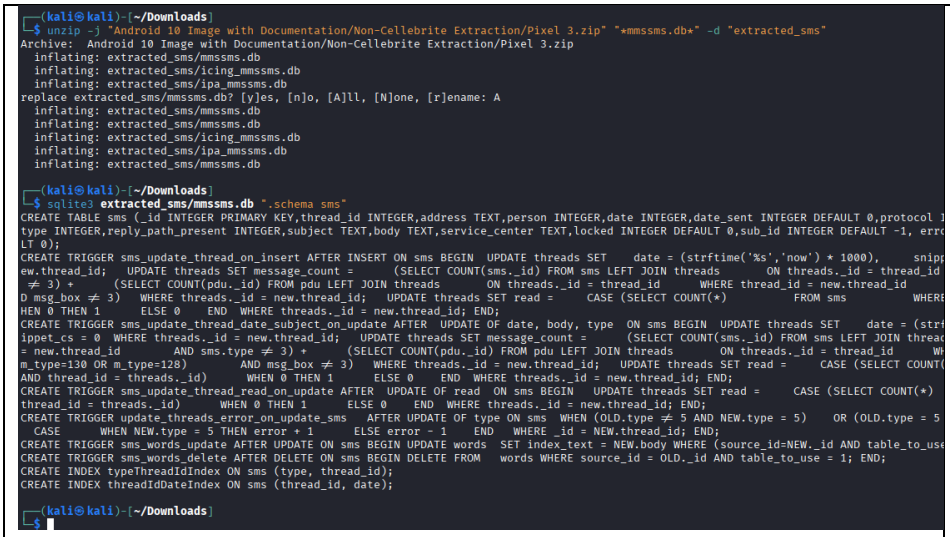
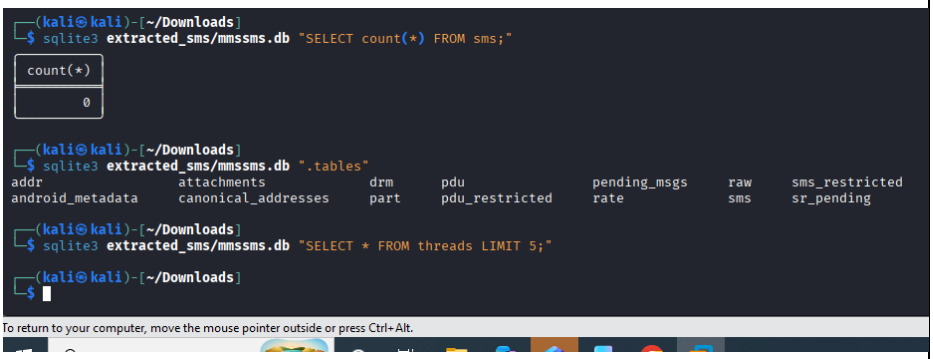
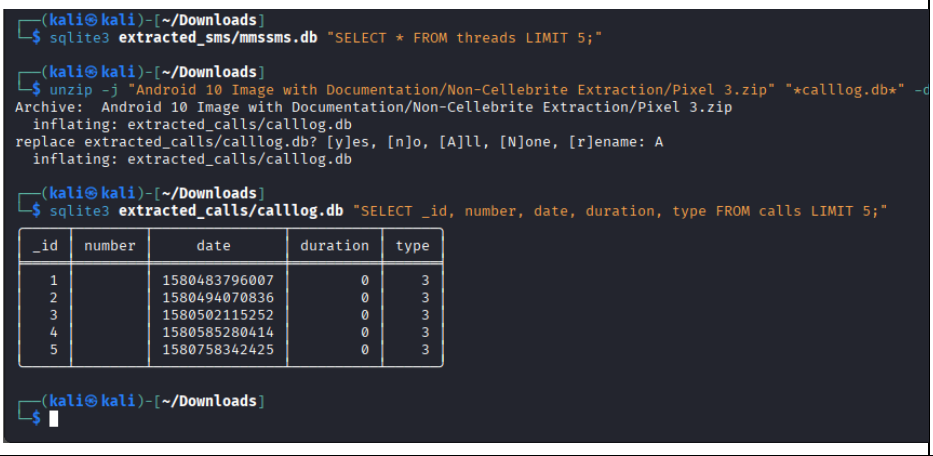


```
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  inflating: extracted_users/journal
  inflating: extracted_users/0.xml
  inflating: extracted_users/appwidgets.xml
  inflating: extracted_users/app_idle_stats.xml
  inflating: extracted_users/package-restrictions.xml
  inflating: extracted_users/photo.png
  inflating: extracted_users/android.accounts.AccountAuthenticator.xml
  inflating: extracted_users/android.content.SyncAdapter.xml
  inflating: extracted_users/roles.xml
  inflating: extracted_users/runtime-permissions.xml
  inflating: extracted_users/settings_config.xml
  inflating: extracted_users/settings_global.xml
  inflating: extracted_users/settings_secure.xml
  inflating: extracted_users/settings_ssaid.xml
  inflating: extracted_users/settings_system.xml
  inflating: extracted_users/wallpaper
  inflating: extracted_users/wallpaper_info.xml
  inflating: extracted_users/wallpaper_orig
  inflating: extracted_users/11.xml
  inflating: extracted_users/appwidgets.xml
  inflating: extracted_users/app_idle_stats.xml
  inflating: extracted_users/device_policies.xml
  inflating: extracted_users/package-restrictions.xml
  inflating: extracted_users/photo.png
  inflating: extracted_users/android.accounts.AccountAuthenticator.xml
  inflating: extracted_users/android.content.SyncAdapter.xml
  inflating: extracted_users/roles.xml
  inflating: extracted_users/runtime-permissions.xml
  inflating: extracted_users/settings_secure.xml
  inflating: extracted_users/settings_ssaid.xml
  inflating: extracted_users/settings_system.xml
  inflating: extracted_users/wallpaper
  inflating: extracted_users/wallpaper_info.xml
  inflating: extracted_users/wallpaper_orig
  inflating: extracted_users/userlist.xml

┌──(kali㉿kali)-[~/Downloads]
└─$
```

To return to your computer, move the mouse pointer outside or press Ctrl+Alt

SS-05	Telephony database schema and counts	Database path, table and query

		
SS-06 	SMS conversation query result	Record IDs, raw timestamp fields and source database
SS-07 	MMS pdu/part relationship	PDU/part IDs and attachment/text reference
SS-08	Google Messages comparison	Matched records and matching criteria

<table>
<tr>
<td>

```
┌──(kali㉿kali)-[~/Downloads]
└─$ unzip -j "Android 10 Image with Documentation/Non-Cellebrite Extraction/Pixel 3.zip" "*packages.xml*" -d "extracted_packa
Archive:  Android 10 Image with Documentation/Non-Cellebrite Extraction/Pixel 3.zip
  inflating: extracted_packages/packages.xml
replace extracted_packages/packages.xml? [y]es, [n]o, [A]ll, [N]one, [r]ename: A
  inflating: extracted_packages/packages.xml

┌──(kali㉿kali)-[~/Downloads]
└─$ head -n 20 extracted_packages/packages.xml
<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<packages>
    <version sdkVersion="29" databaseVersion="3" fingerprint="google/blueline/blueline:10/QQ1A.200105.003/6079943:user/releas
    <version volumeUuid="primary_physical" sdkVersion="29" databaseVersion="3" fingerprint="google/blueline/blueline:10/QQ1A.
    <permission-trees>
        <item name="com.google.android.googleapps.permission.GOOGLE_AUTH" package="com.google.android.gsf" />
    </permission-trees>
    <permissions>
        <item name="com.google.android.apps.nexuslauncher.permission.READ_SETTINGS" package="com.google.android.apps.nexuslau
        <item name="com.google.android.settings.intelligence.apiproxy" package="com.google.android.settings.intelligence" pro
        <item name="com.google.android.gms.auth.api.phone.permission.SEND" package="com.google.android.gms" protection="2" />
        <item name="android.permission.REAL_GET_TASKS" package="android" protection="18" />
        <item name="android.permission.ACCESS_CACHE_FILESYSTEM" package="android" protection="18" />
        <item name="android.permission.REMOTE_AUDIO_PLAYBACK" package="android" protection="2" />
        <item name="android.permission.DOWNLOAD_WITHOUT_NOTIFICATION" package="com.android.providers.downloads" />
        <item name="com.google.android.googlequicksearchbox.permission.LENSVIEW_BROADCAST" package="com.google.android.google
        <item name="com.google.android.apps.dreamliner.permission.WRITE_DREAMLINER_INFO" package="com.google.android.apps.dre
        <item name="com.google.android.apps.photos.permission.C2D_MESSAGE" package="com.google.android.apps.photos" protectio
        <item name="android.permission.MANAGE_APPOPS" package="android" protection="2" />
        <item name="android.permission.REGISTER_WINDOW_MANAGER_LISTENERS" package="android" protection="2" />

┌──(kali㉿kali)-[~/Downloads]
└─$
```

</td>
<td></td>
<td></td>
</tr>
<tr>
<td>

SS-09

```
┌──(kali㉿kali)-[~/Downloads]
└─$ unzip -j "Android 10 Image with Documentation/Non-Cellebrite Extraction/Pixel 3.zip" "*WifiConfigStore
Archive:  Android 10 Image with Documentation/Non-Cellebrite Extraction/Pixel 3.zip
  inflating: extracted_wifi/WifiConfigStore.xml
replace extracted_wifi/WifiConfigStore.xml? [y]es, [n]o, [A]ll, [N]one, [r]ename: A
  inflating: extracted_wifi/WifiConfigStore.xml
  inflating: extracted_wifi/WifiConfigStore.xml
  inflating: extracted_wifi/WifiConfigStore.xml

┌──(kali㉿kali)-[~/Downloads]
└─$ head -n 25 extracted_wifi/WifiConfigStore.xml
<?xml version='1.0' encoding='utf-8' standalone='yes' ?>
<WifiConfigStoreData>
<int name="Version" value="2" />
<Integrity>
<byte-array name="EncryptedData" num="48">281d2b9fdac123760dbbfb0b87216e42500790ed47f319699fc68e73c29af11bl
<byte-array name="IV" num="12">b6722d81a5ed924b76c5c793</byte-array>
</Integrity>
<NetworkList />
<DeletedEphemeralSSIDList>
<map name="SSIDList" />
</DeletedEphemeralSSIDList>
<PasspointConfigData>
<ProviderList />
</PasspointConfigData>
<OpenNetworkNotifierBlacklistConfigData>
<set name="SSIDSet">
<string>EX6100_NETGEAR_2GEXT</string>
</set>
</OpenNetworkNotifierBlacklistConfigData>
<CarrierNetworkNotifierBlacklistConfigData />
<NetworkRequestMap />
<WakeupConfigStoreData>
<FeatureState>
<boolean name="IsActive" value="false" />
<boolean name="IsOnboarded" value="false" />

┌──(kali㉿kali)-[~/Downloads]
└─$
```

</td>
<td>WhatsApp or second-app messages</td>
<td>Database, table, record IDs and status interpretation source</td>
</tr>
<tr>
<td>

SS-10

```
┌──(kali㉿kali)-[~/Downloads]
└─$ unzip -j "Android 10 Image with Documentation/Non-Cellebrite Extraction/Pixel 3.zip" "*History*" -d "extracted_browser"
Archive:  Android 10 Image with Documentation/Non-Cellebrite Extraction/Pixel 3.zip
  inflating: extracted_browser/product@app@LocationHistoryPrebuilt@LocationHistoryPrebuilt.apk@classes.art
  inflating: extracted_browser/product@app@LocationHistoryPrebuilt@LocationHistoryPrebuilt.apk@classes.vdex
  inflating: extracted_browser/History
  inflating: extracted_browser/UserHistory.en_US.dict
 extracting: extracted_browser/notiInvokeHistory.xml
  inflating: extracted_browser/LocationHistoryPrebuilt-16
  inflating: extracted_browser/LocationHistoryPrebuilt.vdex
replace extracted_browser/product@app@LocationHistoryPrebuilt@LocationHistoryPrebuilt.apk@classes.art? [y]es, [n]o, [A]ll, [N]one, [r]e
  inflating: extracted_browser/product@app@LocationHistoryPrebuilt@LocationHistoryPrebuilt.apk@classes.art
  inflating: extracted_browser/product@app@LocationHistoryPrebuilt@LocationHistoryPrebuilt.apk@classes.vdex
  inflating: extracted_browser/History
  inflating: extracted_browser/UserHistory.en_US.dict
 extracting: extracted_browser/notiInvokeHistory.xml
  inflating: extracted_browser/LocationHistoryPrebuilt-16
  inflating: extracted_browser/LocationHistoryPrebuilt.vdex

┌──(kali㉿kali)-[~/Downloads]
└─$ sqlite3 extracted_browser/History ".tables"
downloads     downloads_slices     downloads_url_chains     keyword_search_terms     meta     segment_usage     segments     typed_url

┌──(kali㉿kali)-[~/Downloads]
└─$
```

</td>
<td>Maps/photo metadata and timeline</td>
<td>Source path, metadata fields, offset status and evidence IDs</td>
</tr>
</table>